Unit: 23 Unit 8: Quadratics **Name:** ____________
Topic: Solving by Factoring and Other Methods **Date:** __________

Challenge (Habanero)

Q1. What is the factored form of $x^2 + 5x + 6$?
(A) $(x+2)(x+3)$
(B) $(x-2)(x-3)$
(C) $(x+2)(x-3)$
(D) $(x-2)(x+3)$
(E) $(x+1)(x+4)$

Q2. Solve $x^2 - 7x + 12 = 0$ by factoring.
(A) $x = 3, 4$
(B) $x = -3, -4$
(C) $x = 3, -4$
(D) $x = -3, 4$
(E) $x = 2, 6$

Q3. A gamer has $x^2 + 2x - 15$ points in a game. Factoring the expression yields...
(A) $(x-3)(x+5)$
(B) $(x+3)(x-5)$
(C) $(x+1)(x-15)$
(D) $(x-1)(x+15)$
(E) $(x+2)(x-7)$

Q4. What values of x satisfy $(x-2)(x+1) = 0$?
(A) $x = -1, 2$
(B) $x = 1, -2$
(C) $x = -1, -2$
(D) $x = 1, 2$
(E) $x = 0, 3$

Q5. A new smartphone's battery life B is given by $B = x^2 - 4x + 4$. Solve for x when $B = 0$.
(A) $x = 2$
(B) $x = -2$
(C) $x = 0, 4$
(D) $x = 1, 3$
(E) $x = 2, -2$

Q6. Solve for y in the equation $y^2 - 9y + 20 = 0$.
(A) $y = 4, 5$
(B) $y = -4, -5$
(C) $y = 4, -5$
(D) $y = -4, 5$
(E) $y = 1, 9$

Q7. A coder wants to solve $x^2 + x - 6 = 0$. The factored form is...
(A) $(x-2)(x+3)$
(B) $(x+2)(x-3)$
(C) $(x+1)(x-6)$
(D) $(x-1)(x+6)$
(E) $(x+2)(x+3)$

Q8. The score S in a game is given by $S = x^2 - 5x + 6$. Solve $S = 0$ by factoring.
(A) $x = 2, 3$
(B) $x = -2, -3$
(C) $x = 2, -3$
(D) $x = -2, 3$
(E) $x = 1, 4$

Open Ended Questions (FRQ)

Q9. Solve the equation $x^2 + 4x + 4 = 0$ by factoring and verify the solutions.

Q10. The cost C of data usage is given by $C = x^2 - 3x - 4$. Find the values of x for which $C = 0$ and explain the significance of these values.

Chef's Tips

- Tip 1: Ensure students verify their solutions.
- Tip 2: Emphasize the importance of factoring in problem-solving.
- Tip 3: Encourage students to use real-world examples like tech and gaming to illustrate quadratic concepts.